BSc (Hons) Artificial Intelligence and Data Science

Module: CM2606 Data Engineering

Individual Coursework Report

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Content

1. Introduction

2. Dataset Selection

3. Technical implementation

4. Pipeline Design

5. Discussion and Conclusion

6. Evidence

- Introduction

In this project, I used the Sample Superstore sales dataset to build a complete ETL pipeline on AWS S3. The process started by uploading the raw file from the local data_source folder to the S3 raw folder using Boto3, which served as the data lake. After that, I applied several transformations using Pandas and Boto3, such as converting date columns to proper format, filling in missing values, removing repeated rows, and adding new calculated fields.

Once the cleaning was finished, the final cleaned data was uploaded to the S3 processed folder as a CSV file. This output is now ready to be used for generating business insights through a profit prediction model.

- Dataset Selection

The selected dataset includes detailed superstore sales transactions with important columns like Order Date, Ship Date, Category, Region, Segment, Sales, Profit, Quantity and Discount. These columns are very useful for studying customer purchase behavior, finding which products give more profit, and understanding sales performance in different areas.

Insight Generation Mechanism

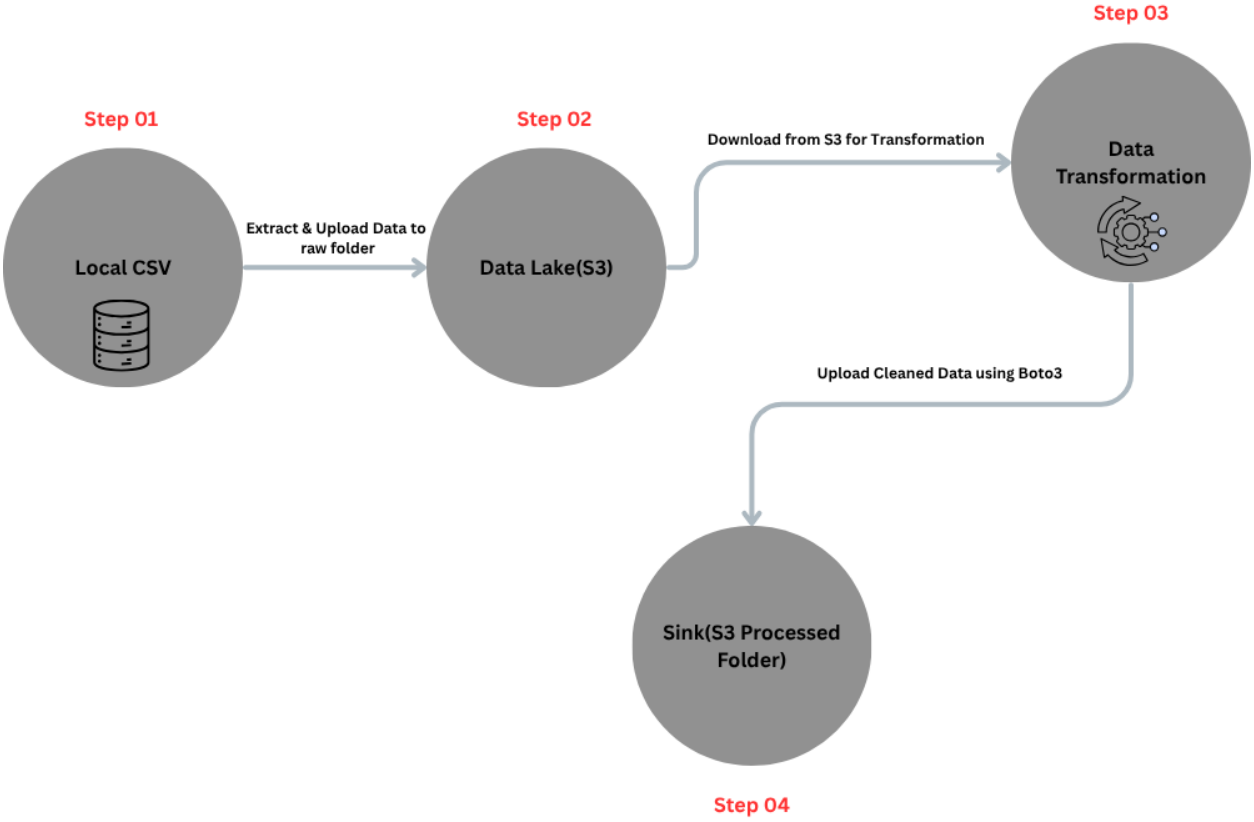
The main goal of this pipeline is to prepare clean data that can support various analytical scenarios for better business understanding, After the ETL process, the data can be used to build a profit prediction model. Possible insights include monthly sales and profit patterns, best-performing product categories, differences in sales between regions, how discounts affect profit, and which customer segments buy the most. These insights will help the superstore make smarter decisions about stock ordering, pricing strategies, and marketing focus.

- Technical Implementation

I chose Python as the main language for the ETL pipeline because it is simple, flexible and has excellent libraries for data handling. Pandas was used for all the data cleaning and processing steps since the dataset size is manageable.

The Boto3 library was used to connect to Amazon S3 for uploading and downloading data. The pipeline was split into three separate Python files: ingestion.py for uploading to S3, etl.py for cleaning and downloading/uploading, and main.py to run everything together. Running a single command (python src/main.py) executes the full pipeline automatically, making the whole process quick and easy to repeat.

- Pipeline Design



• **Data Source**: The original Sample - Superstore.csv file downloaded from Kaggle and placed in the local data_source folder.

• **Data Lake**: Amazon S3 raw folder acts as the temporary storage area for the original data.
• **Ingestion Flow**: The ingestion.py script uses Boto3 to upload the file from data_source to the S3 raw folder.

• **ETL Flow**: The etl.py script downloads the file from S3, performs date formatting, missing value treatment, duplicate removal, feature creation, and monthly aggregation using Pandas, then uploads the result back to S3.

• **Data Storage (Sink)**: Amazon S3 processed folder saves the final clean file (clean_superstore_data.csv) which can be directly used for analysis.

• **Orchestration**: The main.py file runs all steps in one go when executed.

- Discussion and Conclusion

The biggest challenge I faced was handling the CSV encoding error and correctly connecting to S3 using access keys. I solved this by using the encoding="ISO-8859-1" option in Pandas and by creating proper IAM access keys.

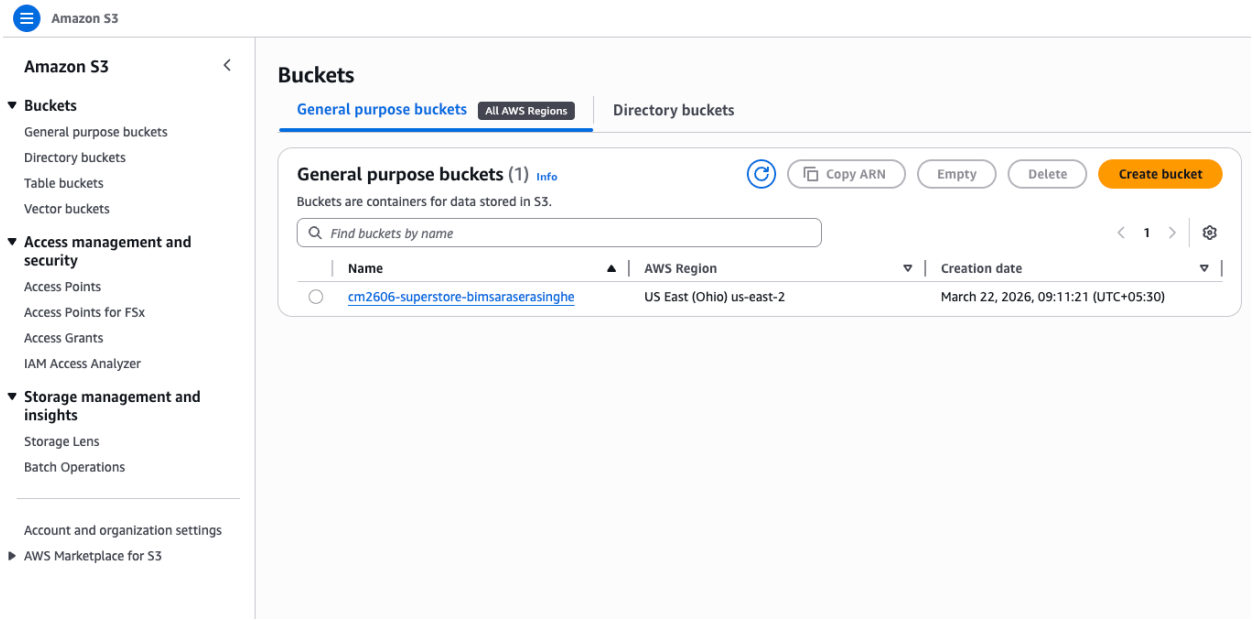
Using AWS S3 with Boto3 made the pipeline cloud-based and closer to real industry practices. This solution fully satisfies all the requirements of the coursework. In future, I can improve it by moving the same pipeline to AWS Glue or connecting the processed data to Redshift and QuickSight for advanced dashboards.

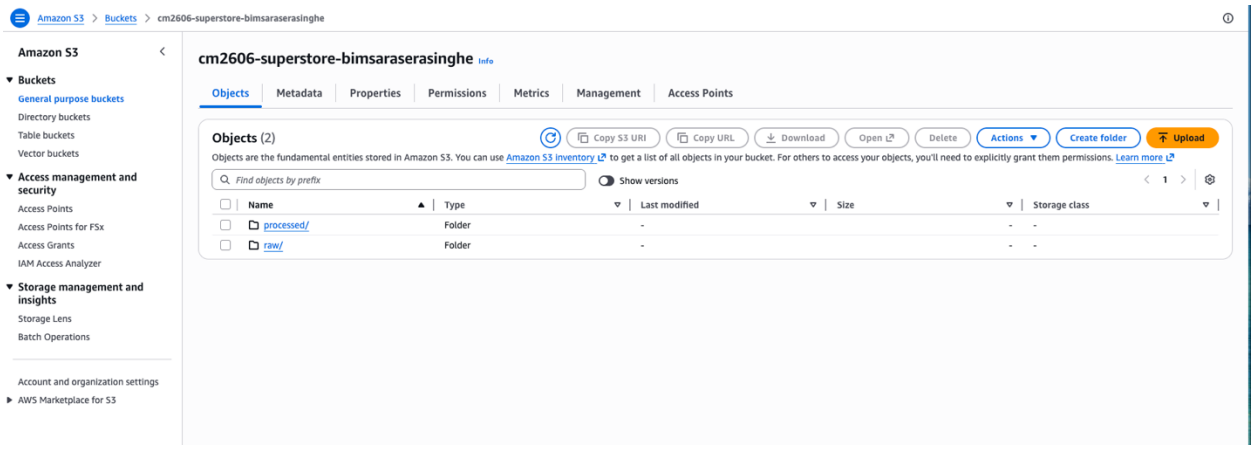
- Evidence

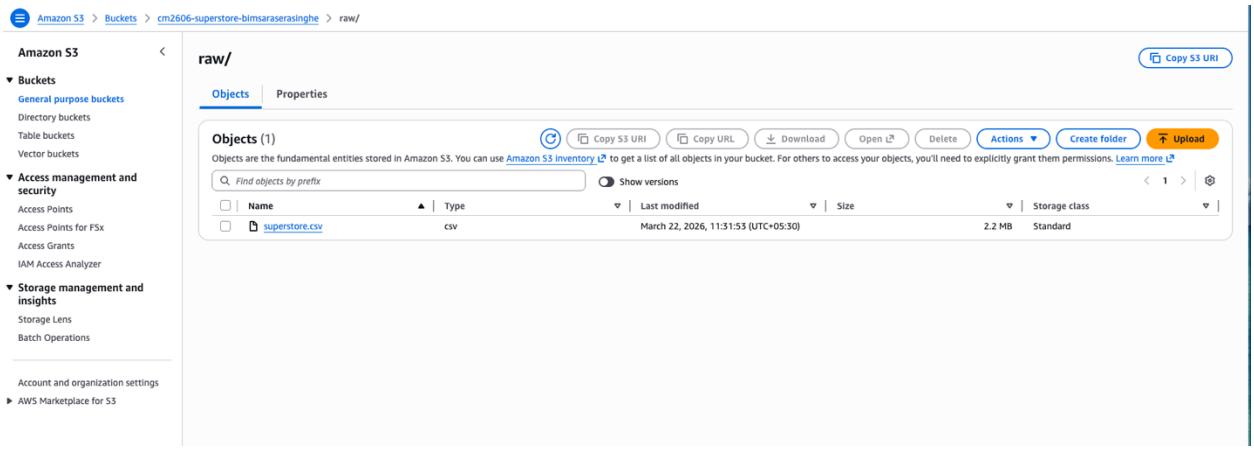
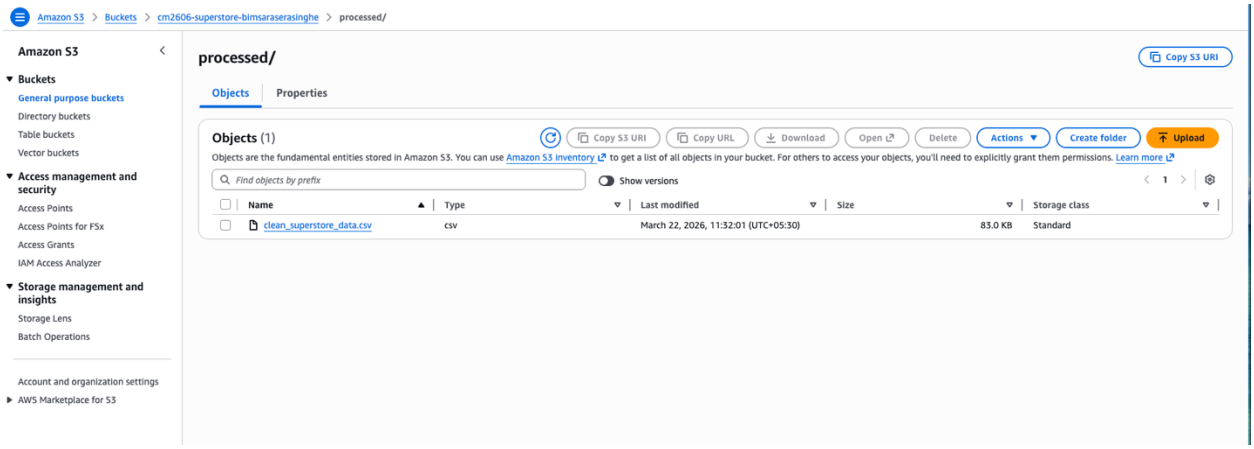
1. Terminal

```
(base) bimsaraserasinghe@Bimsaras-MacBook-Air CM2606_Coursework % python src/main.py
🚀 Starting full pipeline...
Current folder: /Users/bimsaraserasinghe/Downloads/CM2606_Coursework
Copied Sample - Superstore.csv
🎉 Step 1 done! Data is now in raw folder (Data Lake)
Starting cleaning...
Step 2 done! Clean data saved in processed folder (Sink)
🎉 ALL DONE! Your pipeline is complete.
(base) bimsaraserasinghe@Bimsaras-MacBook-Air CM2606_Coursework %
```

2. S3 Raw Folder







3. S3 Processed Folder